Project1:

Evaluating Two-Year HAART Treatment Response Among Hard Drug Users and Non-Users

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Introduction

The data for this project comes from the Multicenter AIDS Cohort Study (MACS), which is an ongoing prospective cohort study of HIV-1 infection in homosexual and bisexual men in four major U.S. cities. The dataset contains up to 8 years of annual follow-up on 715 HIV-infected men after initiating HAART (highly active antiretroviral therapy), including longitudinal laboratory measures, quality-of-life outcomes, and demographic and health information.

The participants were seen annually with year 0 being their last untreated visit, just before starting HAART. All other visits, from year 1 to year 8, are on the treatment plan. Data included four measures of treatment response. The first outcome is viral load defined as the number of HIV RNA copies per milliliter of blood. The second outcome is CD4+ T-cell count, which is a measure of immune health. The last two are quality-of-life (QOL) measures derived from the SF-36 instrument. The aggregate physical health score and the aggregate mental health score, each ranging from 0 to 100, with higher scores indicating better perceived health.

The purpose of this report is to answer the following hypothesis/questions of interest:

1. Year-2 viral load, measured as $\log_{10}$ HIV RNA copies/mL, will be higher among homosexual and bisexual men with HIV-1 who reported baseline hard drug use compared with non-users.
2. Year-2 CD4+ T-cell counts, will be lower among homosexual and bisexual men with HIV-1 who reported baseline hard drug use compared with non-users.
3. Year-2 physical and mental QOL will differ between baseline hard-drug users and non-users.
4. Adherence during follow-up partially mediates the association between baseline hard drug use and Year-2 outcomes.
5. Participants missing Year-2 measurements will differ from those with complete data with respect to baseline hard drug use and baseline outcome measures.

Methods

2.1 Data Cleaning

The original dataset was structured in long format with repeated annual visits per participant. For this analysis, data were reshaped to a subject-level format including baseline (Year 0) and Year 2 measurements. Each row therefore represented a single participant with baseline covariates and Year 2 outcomes.

Baseline hard drug use was coded as a binary indicator (Yes vs. No). Adherence during follow-up was dichotomized (<95% vs. ≥95%). Follow-up status was defined as a binary variable indicating whether participants had complete baseline and Year 2 data required for analysis (Complete vs. Missing).

The race/ethnicity covariate was categorized as White non-Hispanic, Black non-Hispanic, and Hispanic. We acknowledge that race is a social construct and that observed differences likely reflect structural inequities rather than biological differences. Education was categorized as high school or less, college, and post-graduate. Smoking status was categorized as former or current smoker. Because viral load was highly right-skewed, values were $\log_{10}$-transformed prior to modeling. CD4 count and QOL measures were retained on their original scales based on preliminary diagnostic assessment. Implausible BMI values (999, −1, or >100) were coded as missing.

Participants without both baseline and Year 2 measurements were excluded from complete-case regression analyses.

2.2 Data Analysis

Treatment response at Year 2 was evaluated using Bayesian linear regression models for each outcome. Outcomes included $\log_{10}$ viral load (VL), CD4 count, physical QOL, and mental QOL. All models adjusted for baseline hard drug use (primary exposure), baseline value of the outcome, age, BMI, smoking status, education, and race/ethnicity. This approach allows direct evaluation of differences between hard drug users and non-users while adjusting for potential confounders and estimating adjusted mean differences in treatment response.

Models were estimated using the brms package in R, which fits Bayesian models via Stan using Hamiltonian Monte Carlo with the No-U-Turn sampler (HMC-NUTS). A Gaussian likelihood was specified for all outcomes. Weakly informative priors were assigned to regression coefficients and intercepts using normal distributions with mean 0 and large variance [N(0, 100^2)]. Residual variance was assigned a half-Student-t prior with 3 degrees of freedom to allow for heavier tails [Half-t(3,0, 100^2)].

Four Markov chains were run with 2,000 iterations each, including 1,000 warmup iterations. No initial values were specified, so the values were drawn uniformly from the interval (-2,2). Convergence was assessed using the Gelman–Rubin diagnostic ($\hat{R} < 1.01$) and adequate effective sample size (ESS) should be greater than 1000 (Bürkner, 2017). For each outcome, the posterior median and 95% highest density interval (HDI), the probability of direction in the hypothesized direction, the posterior probability of exceeding absolute clinical threshold (Pr(|Outcome|>threshold)), and WAIC differences were reported (SEXIT framework).

Frequentist linear regression models with identical covariate structure were also fit for comparison. Two-sided tests with $\alpha = 0.05$ were used to determine statistical significance.

Because Bayesian inference is based on posterior probability distributions rather than repeated null hypothesis significance testing, formal adjustments for multiple comparisons were not performed. All analyses were conducted using R (version 4.5.1 “Great Square Root”).

2.2.1 Mediation

To evaluate whether adherence mediated the association between baseline hard drug use and Year 2 outcomes, two models were fit for each outcome: (1) a total-effect model excluding adherence, and (2) a direct-effect model including adherence. Evidence consistent with mediation was assessed by examining attenuation of the hard drug use coefficient after inclusion of adherence. The proportion mediated was calculated as $(TE - DE) / TE$. This approach represents an exploratory difference-in-coefficients method and does not constitute a formal causal mediation analysis.

2.2.2 Loss to Follow-up

Participants without both baseline and Year 2 measurements were excluded from complete-case regression analyses. Follow-up status was retained as a separate binary variable to evaluate potential bias due to loss to follow-up. Baseline characteristics were compared between participants retained and those lost to follow-up using Wilcoxon rank-sum tests for continuous variables. The Wilcoxon test was chosen because the outcome variables were non-normally distributed and exhibited skewness, making a non-parametric approach more appropriate than parametric t-tests.

Results

Characteristic	N	Overall N = 463[1,2]	Hard Druge Use: No N = 427[1,2]	Hard Druge Use: Yes N = 36[1,2]
Mental QOL	463			
Mean (SD)		44.9 (13.6)	45.2 (13.8)	42.0 (11.6)
Median (Q1, Q3)		49.1 (36.0, 56.2)	49.5 (36.4, 56.3)	44.4 (30.6, 51.7)
Min - Max		7.2 - 66.0	7.2 - 66.0	22.5 - 59.6
Physical QOL	463			
Mean (SD)		51.2 (8.9)	51.4 (9.0)	49.2 (7.1)
Median (Q1, Q3)		53.5 (46.1, 57.0)	53.7 (47.0, 57.1)	47.4 (44.3, 54.5)
Min - Max		19.2 - 69.0	19.2 - 69.0	31.4 - 62.9
Viral Load (log_{10} copies/mL)	463			
Mean (SD)		4.5 (0.9)	4.5 (0.9)	4.6 (0.9)
Median (Q1, Q3)		4.5 (4.0, 5.1)	4.5 (4.0, 5.1)	4.5 (4.1, 5.5)
Min - Max		0.2 - 8.3	0.2 - 8.3	2.9 - 6.4
CD4 T-cell Count	463			
Mean (SD)		373.3 (202.0)	374.5 (202.3)	360.0 (200.8)
Median (Q1, Q3)		359.8 (228.6, 495.7)	359.8 (234.9, 488.7)	396.7 (187.8, 557.8)
Min - Max		10.9 - 1,217.5	10.9 - 1,217.5	10.9 - 650.2
Age	463			
Mean (SD)		43.2 (8.7)	43.1 (8.7)	43.9 (9.5)
Median (Q1, Q3)		43.0 (37.0, 49.0)	43.0 (37.0, 48.0)	45.5 (36.5, 52.0)
Min - Max		20.0 - 73.0	20.0 - 73.0	29.0 - 61.0
BMI	463			
Mean (SD)		25.2 (4.3)	25.3 (4.4)	23.6 (3.4)
Median (Q1, Q3)		24.6 (22.4, 27.7)	24.8 (22.6, 27.9)	23.3 (21.0, 25.1)
Min - Max		16.5 - 45.3	16.5 - 45.3	18.0 - 31.2
Smoke	463			
Former		287 (62%)	278 (65%)	9 (25%)
Current		176 (38%)	149 (35%)	27 (75%)
Education	463			
College degree or higher		201 (43%)	191 (45%)	10 (28%)
Didn't finish college or lower		262 (57%)	236 (55%)	26 (72%)
Race	463			
Black non-Hisp		122 (26%)	108 (25%)	14 (39%)
Hispanic		45 (9.7%)	42 (9.8%)	3 (8.3%)
White non-Hisp		296 (64%)	277 (65%)	19 (53%)

[1] n (%)

[2] Table includes only subjects that have both baseline and year 2 measurements

Table 1. *Baseline characteristics of HIV-infected men initiating HAART, overall and stratified by baseline hard drug use. This table represents the total number of subjects that will be used in all subsequent analyses.*

Of 715 participants, 463 (65%) had complete Year 2 outcome data and were included in regression analyses. Baseline hard drug users reported lower physical and mental QOL scores and lower CD4 counts compared with non-users. Viral load, age, BMI, and race/ethnicity distributions were similar between groups.

3.1 Treatment Response 2 years After Initiating HAART

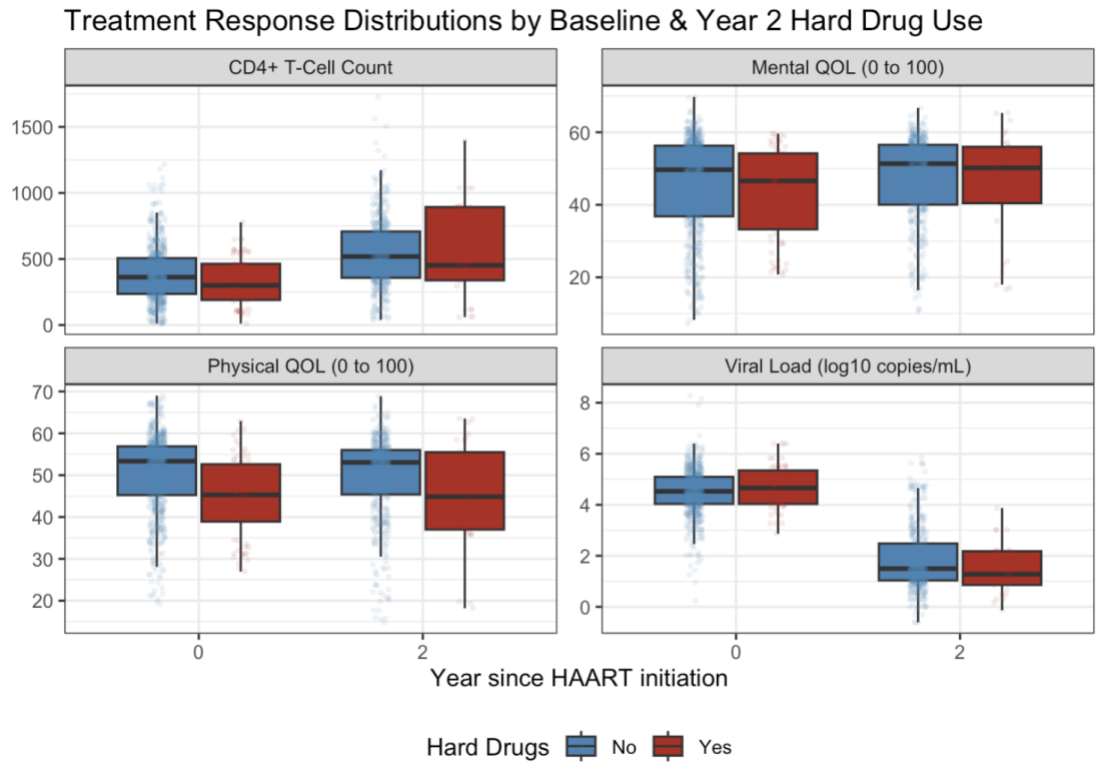

Figure 1. *Distribution of treatment response measures at baseline (Year 0) and Year 2 following HAART initiation, stratified by baseline hard drug use. Boxplots display the median (black line) and interquartile range, with points representing individual observations.*

CD4+ T-cell counts increased from baseline to Year 2 in both groups. Participants reporting hard drug use at baseline generally exhibited slightly lower CD4 counts at both time points but with more variability at year 2. Viral load ($\log_{10}$ copies/mL) declined substantially following treatment initiation in both groups. No apparent differences in viral load are visible at both time

points. Physical and mental quality-of-life scores remained relatively stable over time with drug users typically reporting lower QOL scores across both time points.

	Bayesian				Frequentist			
	N	Users vs Non[1]	95% HDI[2]	WAIC Diff (Full-Red)[3]	N	Users vs Non[1]	95% CI[2]	P-value
CD4 *Mean Diff (cells/mm³)*	463	−163.8	(-231.8, -102.7)	−23.6	463	−164.1	(-226.6, -101.6)	<0.001
$\log_{10}$ VL *Mean Diff (copies/mL)*	463	−0.1	(-0.5, 0.4)	2.4	463	−0.1	(-0.5, 0.3)	0.753
Physical QOL *Mean Diff (0-100 scale)*	463	−3.4	(-6.0, -0.6)	−3.4	463	−3.3	(-6.0, -0.6)	0.017
Mental QOL *Mean Diff (0-100 scale)*	463	−0.3	(-3.7, 3.2)	3.3	463	−0.3	(-3.8, 3.2)	0.865

[1] For hard drug users versus non-users, estimates represent adjusted differences (Users – Non-users). Negative values indicate lower outcome values among users.

[2] Bayesian intervals are 95% high density intervals; Frequentist intervals are 95% confidence intervals.

[3] For WAIC, lower values indicate better model fit. WAIC difference is the full model minus model w/out exposure (hard drug use).

Table 2. *Baseline and covariate adjusted differences in Year-2 treatment response between baseline hard drug users and non-users under Bayesian and frequentist regression frameworks. Estimates represent mean differences for each outcome. Bayesian highest density intervals (HDI) and frequentist confidence intervals (CI) are reported alongside model-based estimates.*

Baseline hard drug use was associated with significantly lower CD4 counts at Year 2. The posterior median adjusted difference was −164 cells/mm³ (95% HDI: −263 to −103; $\hat{R}$=1.00, ESS=6929), with posterior probability of a negative effect (Pr(β<0)] was approximately 100%. The probability that this exceeded the prespecified clinically meaningful threshold of 50 cells/mm³ was 99.5%.

No meaningful difference in viral load was observed (P(β>0)=49.3%)). The posterior median difference was approximately −0.1 log_{10} copies/mL (95% HDI: −0.5 to 0.4; $\hat{R}$=1.00, ESS=8267), and the probability of exceeding the clinical threshold of 0.5 log_{10} copies/mL was 2.3%.

Hard drug use was associated with modestly lower physical QOL at Year 2 (−3.3 points; 95% HDI: −6.0 to −0.6; $\hat{R}$=1.00, ESS=8267). The posterior probability that this exceeded the prespecified 2-point clinical threshold was 83%. No evidence of association was observed for mental QOL, and posterior probability of exceeding 2-points was 26%.

Differences in WAIC greater than three are generally considered strong evidence of a better fitting model. The large WAIC reduction for CD4 indicates meaningful added predictive value of hard drug use, whereas WAIC differences for other outcomes were smaller and provided weaker evidence of better fit.

Frequentist estimates were consistent in effect sizes and direction with Bayesian results.

3.2 Mediation

	Bayesian						Frequentist			
	N	W/ADH[1]	W/out ADH	Prop Med[2]	95% HDI[3]	Pr(Med>0)	N	W/ADH[1]	W/out ADH	Prop Med
CD4 *Mean Diff (cells/mm³)*	463	−170.0	−164.0	−0.037	(-0.8, 0.4)	0.449	463	−169.9	−164.1	−0.035
$\log_{10}$ VL *Mean Diff (copies/mL)*	463	1.0	0.9	−0.109	(-2.9, 0.7)	0.434	463	1.0	0.9	−0.113
Physical QOL *Mean Diff (0-100 scale)*	463	−3.6	−3.3	−0.078	(-3.7, 0.8)	0.455	463	−3.6	−3.3	−0.070
Mental QOL *Mean Diff (0-100 scale)*	463	−0.5	−0.3	−0.874	(-12.0, 15.1)	0.738	463	−0.5	−0.3	−0.806

[1] For hard drug users versus non-users, estimates represent adjusted differences (Users – Non-users). Negative values indicate lower outcome values among users.

[2] Proportion Mediated = (Model with ADH - Model without ADH) / Model without ADH or (TE-DE)/TE

[3] 95% high density intervals for proportion mediated

***Table 3**. Association of hard drug use with clinical and quality-of-life outcomes, with and without adjustment for adherence (ADH), and estimated proportion mediated.*

For all outcomes, inclusion of adherence slightly increased the magnitude of the hard drug use coefficient, resulting in negative proportions mediated. Bayesian posterior intervals for the

proportion mediated included zero across outcomes, and posterior probabilities that the mediated proportion was positive were near 0.5, except for mental QOL. These findings indicate inconsistent mediation (suppression) and provide little evidence that adherence mediates the relationship between baseline hard drug use and treatment outcomes.

3.3 Loss to Follow-up

	Non-users				Users			
Characteristic	**N**	**Complete** N = 427	**Missing** N = 222	**p-value**[1]	**N**	**Complete** N = 36	**Missing** N = 30	**p-value**[2]
Baseline viral load (log10 copies/mL)	610			0.571	63			0.509
Mean (SD)		4.5 (0.9)	4.4 (0.9)			4.6 (0.9)	4.8 (0.8)	
Median (Q1, Q3)		4.5 (4.0, 5.1)	4.5 (4.0, 5.0)			4.5 (4.1, 5.5)	4.8 (4.0, 5.2)	
Missing Both Measurements		0 (0%)	39 (18%)			0 (0%)	3 (10%)	
Baseline CD4 T-cell Count	625			0.172	66			0.216
Mean (SD)		374.5 (202.3)	407.9 (228.5)			360.0 (200.8)	305.1 (167.3)	
Median (Q1, Q3)		359.8 (234.9, 488.7)	369.9 (236.1, 537.8)			396.7 (187.8, 557.8)	258.3 (203.9, 446.2)	
Missing Both Measurements		0 (0%)	24 (11%)					
Baseline physical QOL	647			0.021	66			<0.001
Mean (SD)		51.4 (9.0)	49.3 (9.9)			49.2 (7.1)	39.5 (9.6)	
Median (Q1, Q3)		53.7 (47.0, 57.1)	52.7 (42.8, 56.5)			47.4 (44.3, 54.5)	36.7 (31.2, 50.1)	
Missing Both Measurements		0 (0%)	2 (0.9%)					
Baseline mental QOL	647			0.496	66			0.327
Mean (SD)		45.2 (13.8)	46.5 (12.5)			42.0 (11.6)	45.0 (13.2)	
Median (Q1, Q3)		49.5 (36.4, 56.3)	49.7 (37.8, 56.2)			44.4 (30.6, 51.7)	48.0 (37.3, 56.0)	
Missing Both Measurements		0 (0%)	2 (0.9%)					

[1] Wilcoxon rank sum test

[2] Wilcoxon rank sum test; Wilcoxon rank sum exact test

***Table 4**. Baseline clinical and quality-of-life measures between participants by follow-up status, stratified by baseline hard drug use. Participants with complete baseline and Year 2 data (n = 463) and those lost to follow-up (n = 252) out of 715 total participants*

Within both non-users and users of hard drugs at baseline, participants lost to follow-up had significantly lower baseline physical QOL compared with those with both measurements (non-users: $p = 0.021$; users: $p < 0.001$). No significant differences in baseline viral load, CD4 count, or mental QOL were observed between complete and missing participants within either stratum. These results suggest that attrition was associated with poorer baseline physical health but not with baseline laboratory measures.

Discussion

In this cohort of HIV-infected men initiating HAART, baseline hard drug use was associated with substantially lower CD4 T-cell counts and modestly lower physical quality-of-life scores at Year 2. No meaningful differences were observed in viral load or mental QOL. These findings suggest that while immunologic recovery and perceived physical health differences persist two years after treatment initiation, viral load suppression and mental health does not differ meaningfully by baseline hard drug use.

Exploratory mediation analyses provided little support for adherence as a mediator of these associations. In fact, the direct effects were slightly larger in magnitude than the total effects, indicating inconsistent mediation. Because this approach only looked at differences in the hard drug coefficient estimates, it does not formally estimate natural indirect effects and relies on strong assumptions regarding confounding, these findings should be interpreted cautiously.

Loss to follow-up was associated with lower baseline physical quality-of-life scores within both hard drug users and non-users but was not associated with baseline laboratory measures. These findings indicate that missingness was unlikely to be completely at random (MCAR) and may instead be consistent with missing-at-random (MAR) mechanism, where missingness is dependent on observed variables, possibly baseline physical health. Because baseline physical QOL was included as a covariate in all models, the potential for bias due to differences in dropout probabilities between users and non-users is reduced but cannot be entirely ruled out.

4.1 Limitations

Several limitations should be noted. First, the observational design of this study can suffer from bias due to exposure of interest not being randomly assigned and unmeasured confounders,

which may lead to misleading associations. Second, adherence was measured as a binary variable (<95% vs ≥95%) and may be too simplistic for proper classification. Because of this, mediation was evaluated using an exploratory attenuation approach rather than a formal causal mediation framework. There was a relatively small number of baseline hard drug users (36), which may not contain enough power to accurately make comparisons.

Lastly, the interpretation of SF-36 physical and mental QOL scores is limited because meaningful clinical interpretation typically requires comparison to age-, sex-, and population-specific normative data. Scores below the 25th percentile of normative distributions are often considered indicative of significant impairment; however, such normative benchmarking was not incorporated in the present analysis.

4.2 Conclusions

Overall, these findings suggest that baseline hard drug use is associated with poorer immunologic recovery and lower physical quality of life following HAART initiation, but adherence does not appear to meaningfully mediate these associations.

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ChatGPT was used to check R code errors for some of the packages used and to grammar check (OpenAI, 2025).

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Reproducibility Statement

The statistical code used in this analysis is available on my GitHub under the Code directory at: https://github.com/rysummers/BIOS6624-Class/tree/main/Project1

Raw data are not tracked in this repository and is available upon request.

Link to data on local drive: /Users/ryan_summers/GitHub/BIOS6624-Class/Project1/DataRaw/hiv_6624_final.csv